

UNIT III: CCE2 — MODEL ANSWER KEY

Machine Learning Fundamentals & Supervised Learning

Course Pattern: TE IT - 2024 Pattern (SPPU)

Academic Term: Semester V (2026)

Document Type: CCE2 Model Answer Key

Course Code: PCC-302-IT

NOTE:

Full step-by-step worked solutions are provided below. Award **proportional marks** for intermediate steps in numerical problems. Accept alternative valid technical definitions for theory sub-parts. Solutions are structured in the **SPPU Model Answer Key** point-wise format.

Q1. House Price Prediction using Linear Regression — 7 Marks

SCENARIO:

A real-estate company wants to develop a machine learning model to predict house prices based on **area** and **number of bedrooms**.

Part (a) — Regression or Classification? (1 Mark)

- This is a **Regression** problem.
- **Reason:** The target variable (house price) is a **continuous numerical quantity**. Regression models predict a real-valued output, whereas classification predicts a discrete category label.

Part (b) — Two Preprocessing Steps (2 Marks)

1. **Handling Missing Values:** Any missing entries in area or bedroom count must be imputed using the **mean** (for symmetric distributions) or **median** (for skewed data) to prevent model training errors.
2. **Feature Scaling (Normalization / Standardization):** The numerical ranges of area (e.g., hundreds to thousands of sq ft) and bedrooms (e.g., 1–5) differ vastly. Applying **Min-Max Normalization** or **Z-score Standardization** brings both features to a comparable scale so that neither dominates the regression gradient.

Part (c) — Calculate Coefficients and Regression Equation (2 Marks)**Given values:**

$$\bar{x} = 1500, \quad \bar{y} = 32, \quad \sum (x - \bar{x})(y - \bar{y}) = 8400, \quad \sum (x - \bar{x})^2 = 6000$$

Step 1 — Slope (β_1):

$$\beta_1 = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2} = \frac{8400}{6000} = 1.4$$

Step 2 — Intercept (β_0):

$$\beta_0 = \bar{y} - \beta_1 \times \bar{x} = 32 - (1.4 \times 1500) = 32 - 2100 = -2068$$

Regression Equation:

$$\hat{y} = \beta_0 + \beta_1 x = -2068 + 1.4x$$

Part (d) — Regression Evaluation Measure (2 Marks)

- **Mean Squared Error (MSE):** Measures the average squared difference between predicted and actual values. Lower MSE indicates a better-fitting model.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- **R-squared (R^2) — Coefficient of Determination:** Measures the proportion of variance in the dependent variable explained by the model. $R^2 = 1$ is a perfect fit; R^2 closer to 1 indicates better model performance.
- These metrics together give both **magnitude of error** and **relative fit quality**.

Q2. Spam Email Classification — 7 Marks**SCENARIO:**

An email service provider wants to automatically classify incoming emails as **Spam** or **Not Spam**.

Part (a) — Type of Problem and Algorithms (1 Mark)

- Type: **Supervised Binary Classification**
- Suitable Algorithms: **Naive Bayes** and **Logistic Regression** (also acceptable: Decision Tree, SVM)

Part (b) — Naive Bayes for Spam Classification (2 Marks)

1. **Feature Extraction:** Each word (or token) in the email is treated as an independent feature.
2. **Training:** The classifier learns $P(\text{Spam})$, $P(\text{Not Spam})$, and the conditional word probabilities $P(\text{word} | \text{Spam})$ and $P(\text{word} | \text{Not Spam})$ from the training corpus.
3. **Classification:** Using Bayes' Theorem with the **conditional independence assumption**, it computes the posterior probability for each class and assigns the email to the class with the **highest posterior probability**.

Part (c) — Calculate Accuracy, Precision, and Recall (3 Marks)

Given: TP = 80, TN = 90, FP = 10, FN = 20. Total = 80 + 90 + 10 + 20 = 200

Metric	Formula & Calculation	Result
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN} = \frac{170}{200}$	85.00%
Precision	$\frac{TP}{TP+FP} = \frac{80}{90}$	88.89%
Recall	$\frac{TP}{TP+FN} = \frac{80}{100}$	80.00%

Part (d) — Minimizing Missed Spam (1 Mark)

- **Recall** should receive greater attention.
- **Reason:** Missed spam = **False Negative**. Recall = $\frac{TP}{TP+FN}$. Maximizing Recall directly minimizes FN, ensuring the fewest actual spam emails slip through undetected.

Q3. Medical Diagnosis using KNN — 8 Marks**SCENARIO:**

A hospital classifies a new patient (4, 6) as **High Risk** or **Low Risk** using KNN on the training set below.

Patient	Age	Cholesterol	Class
P1	2	3	Low
P2	3	4	Low
P3	5	6	High
P4	6	7	High
P5	4	5	Low
P_new	4	6	?

Table 3.1: Normalized Training Dataset (Age, Cholesterol)

Part (a) — Why Feature Scaling is Important for KNN (2 Marks)

1. KNN computes **Euclidean distance** between data points to identify neighbors.
2. If features have different magnitudes (e.g., Age: 1–100, Annual Income: 10,000–1,00,000), the feature with the **larger numerical range artificially dominates** the distance metric.
3. Feature scaling (e.g., Min-Max Normalization or Z-score Standardization) brings all features to a **comparable range**, ensuring each contributes **equally** to the distance calculation and preventing biased neighbor selection.

Part (b) — Euclidean Distances from New Patient (4, 6) (3 Marks)

Distance formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Patient	Calculation	Distance	Class
P1 (2,3)	$\sqrt{(4-2)^2 + (6-3)^2} = \sqrt{4+9} = \sqrt{13}$	≈ 3.61	Low
P2 (3,4)	$\sqrt{(4-3)^2 + (6-4)^2} = \sqrt{1+4} = \sqrt{5}$	≈ 2.24	Low
P3 (5,6)	$\sqrt{(4-5)^2 + (6-6)^2} = \sqrt{1+0} = 1$	$= 1.00$	High
P4 (6,7)	$\sqrt{(4-6)^2 + (6-7)^2} = \sqrt{4+1} = \sqrt{5}$	≈ 2.24	High
P5 (4,5)	$\sqrt{(4-4)^2 + (6-5)^2} = \sqrt{0+1} = 1$	$= 1.00$	Low

Part (c) — Predicted Class for k = 3 (2 Marks)

Distances in ascending order: P3 (1.00, High), P5 (1.00, Low), P2 (2.24, Low), P4 (2.24, High)

- 3 nearest neighbors: **P3 (High), P5 (Low), P2 (Low)**
- Majority vote: Low = 2, High = 1
- **Predicted class: Low Risk**

(Note: If P4 is selected over P2 in case of tie, result may be High Risk. A tie-breaking rule such as lower index takes precedence.)

Part (d) — Limitation for Large Datasets (1 Mark)

KNN is a **lazy learner** — it defers all computation to inference time. For very large medical datasets, computing distance from the query point to **every training record** is computationally expensive ($O(n)$ per query), making real-time classification impractical.

Q4. Decision Tree for Loan Approval — 8 Marks**SCENARIO:**

A bank builds a decision tree to classify loan applications as **Approved** or **Rejected** based on income, credit history, and employment status.

Part (a) — Suitability of Decision Tree (1 Mark)

- Decision trees produce **explicit if-then rules** that are human-readable and interpretable.
- Banks must legally justify loan decisions; a decision tree provides a **transparent audit trail**, unlike black-box models.

Part (b) — Entropy and Information Gain (2 Marks)

- **Entropy:** A measure of impurity or uncertainty in a dataset. Defined as:

$$H(S) = - \sum_i p_i \log_2(p_i)$$

A dataset containing only one class has $H = 0$ (pure). Maximum entropy occurs when all classes are equally distributed.

- **Information Gain (IG):** The reduction in entropy after splitting on an attribute A :

$$IG(S, A) = H(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} H(S_v)$$

The attribute with the **highest IG** is chosen for splitting, as it reduces uncertainty the most.

Part (c) — Calculate Entropy of the Dataset (2 Marks)

Given: Approved = 6, Rejected = 4, Total = 10

Probabilities: $p_1 = \frac{6}{10} = 0.6$, $p_2 = \frac{4}{10} = 0.4$

$$H(S) = -(p_1 \log_2 p_1 + p_2 \log_2 p_2)$$

$$H(S) = -(0.6 \times \log_2(0.6) + 0.4 \times \log_2(0.4))$$

$$H(S) = -(0.6 \times (-0.737) + 0.4 \times (-1.322))$$

$$H(S) = -(-0.442 - 0.529) = \mathbf{0.971 \text{ bits}}$$

Part (d) — Information Gain for Root Node Selection (2 Marks)

1. For each candidate attribute (Income, Credit History, Employment Status), compute the weighted average entropy of child nodes after splitting.
2. Calculate $IG = (\text{Parent entropy}) - (\text{Weighted child entropy})$.
3. The attribute with the **maximum Information Gain** (i.e., most entropy reduction) is selected as the **root node**.
4. This process repeats recursively at each node using remaining attributes.

Part (e) — Advantage and Limitation (1 Mark)

- **Advantage:** Requires **no feature scaling**; handles both categorical and numerical attributes; produces interpretable if-then rules directly.
- **Limitation:** Highly susceptible to **overfitting** if grown deep without pruning; small data changes can produce drastically different trees.

Q5. Naive Bayes for Customer Classification — 8 Marks

SCENARIO:

An online shopping platform predicts whether a customer will **Purchase** or **Not Purchase** based on receiving a promotional email.

Email Received	Purchase	Not Purchase
Yes	40	20
No	10	30

Table 5.1: Historical Training Data

Part (a) — Naive Bayes Assumption (2 Marks)

The Naive Bayes classifier makes the “**naive**” **assumption of conditional independence**: it assumes that the effect of each feature (attribute value) on the class label is **independent of all other features**, given the class.

Mathematically: $P(x_1, x_2, \dots, x_n \mid C) = \prod_{i=1}^n P(x_i \mid C)$

This strong independence assumption simplifies computation enormously, even though it rarely holds exactly in practice.

Part (b) — Calculate P(Purchase) and P(Not Purchase) (2 Marks)

Total Purchase = 40 + 10 = 50, Total Not Purchase = 20 + 30 = 50, Grand Total = 100

$$P(\text{Purchase}) = \frac{50}{100} = 0.5$$

$$P(\text{Not Purchase}) = \frac{50}{100} = 0.5$$

Part (c) — Conditional Probabilities and Predicted Class (3 Marks)

New customer: **Email = Yes**

Step 1 — Conditional Probabilities:

$$P(\text{Email=Yes} \mid \text{Purchase}) = \frac{40}{50} = 0.8$$

$$P(\text{Email=Yes} \mid \text{Not Purchase}) = \frac{20}{50} = 0.4$$

Step 2 — Posterior Scores (proportional):

$$\text{Score}(\text{Purchase}) = P(\text{Email=Yes} \mid \text{Purchase}) \times P(\text{Purchase}) = 0.8 \times 0.5 = 0.4$$

$$\text{Score}(\text{Not Purchase}) = P(\text{Email=Yes} \mid \text{Not Purchase}) \times P(\text{Not Purchase}) = 0.4 \times 0.5 = 0.2$$

Step 3 — Prediction: Since $0.4 > 0.2$, the predicted class is **Purchase**.

Part (d) — Practical Application (1 Mark)

Sentiment Analysis: Naive Bayes is widely used to classify social media posts, product reviews, or customer feedback as Positive, Negative, or Neutral based on the presence of opinion words.

Q6. Data Preprocessing in an E-Commerce Dataset — 7 Marks

SCENARIO:

An e-commerce dataset contains Age, Gender, Annual Income, and Purchase Amount with **missing values**, **categorical variables**, and **different numerical ranges**.

Part (a) — Two Data-Quality Problems (1 Mark)

1. **Missing Values:** Incomplete records cause errors during model training and bias results.
2. **Inconsistent Numerical Ranges:** Features like Age (18–60) and Annual Income (10,000–10,00,000) have vastly different scales, biasing distance-based algorithms.

Part (b) — Two Methods for Handling Missing Values (2 Marks)

- **Mean / Median Imputation:** Replace missing numerical values with the column **mean** (for symmetric distributions) or **median** (for skewed distributions). Preserves dataset size without data loss.
- **Mode Imputation:** For categorical columns (e.g., Gender), replace missing values with the most frequently occurring category.

(Also acceptable: Row deletion, KNN imputation)

Part (c) — Feature Scaling vs Encoding (2 Marks)

Parameter	Feature Scaling	Feature Encoding
Purpose	Transform numerical range to uniform scale	Convert categorical text to numeric format
Applied to	Continuous numerical columns	Categorical / nominal columns
Example	Min-Max: Annual Income mapped to $[0, 1]$	One-Hot Encoding: Gender (Male/Female) → binary columns

Part (d) — Why Preprocessing Before Training (2 Marks)

1. **Model Correctness:** Raw data with missing values or incorrect types causes training errors or undefined mathematical operations.
2. **Performance Improvement:** Scaled and cleaned data leads to **faster convergence** (gradient descent), **higher accuracy**, and reduces the risk of **overfitting to noise**.

Q7. Training, Validation and Testing — 7 Marks

SCENARIO:

A startup builds a model to predict whether a customer will **cancel their subscription**.

Part (a) — Purpose of Training Dataset (1 Mark)

The **training dataset** is the primary data used to **fit the model** — allowing the algorithm to learn the underlying patterns, adjust internal parameters (weights/coefficients), and minimize the training loss function.

Part (b) — Role of Validation Data (2 Marks)

1. The validation set is a **held-out subset** used **during** the training phase (after each training epoch or tuning step) to evaluate model performance on data it has not trained on.
2. It is used for **hyperparameter tuning** (e.g., selecting optimal k in KNN, tree depth in Decision Trees) and provides early warning of **overfitting** before the model is finalized.

Part (c) — Why a Separate Testing Dataset is Required (2 Marks)

1. The test set remains **completely unseen** during both training and validation — it is isolated until the final model is ready.
2. It provides an **unbiased, final estimate** of real-world generalization performance, simulating how the model will behave on future customer data.
3. Evaluating on training/validation data would give an **overly optimistic** estimate due to data leakage.

Part (d) — 96% Train Accuracy vs 72% Test Accuracy (2 Marks)

- Problem: **Overfitting**
- **Explanation:** The model has memorized the specific patterns and noise in the training data rather than learning the generalizable underlying signal. It performs extremely well on training data but fails to generalize to unseen test data, resulting in a large **generalization gap** (96% → 72%).
- **Remedy:** Apply regularization, reduce model complexity, use cross-validation, or obtain more training data.

Q8. Logistic Regression for Student Performance — 8 Marks

SCENARIO:

A university predicts whether a student is **Likely to Pass** or **Likely to Fail** using attendance, internal marks, and assignment completion.

Part (a) — Why Logistic Regression? (2 Marks)

1. The output is a **binary categorical label** (Pass / Fail) — Logistic Regression is specifically designed for **binary classification**.
2. It outputs a **probability score** between 0 and 1 (via the sigmoid function), which is then thresholded to produce the final class label. It handles linearly separable decision boundaries efficiently.

Part (b) — Linear Regression vs Logistic Regression (2 Marks)

Parameter	Linear Regression	Logistic Regression
Output	Continuous real value ($-\infty$ to $+\infty$)	Probability bounded in $(0, 1)$
Task	Regression (predict numeric values)	Classification (predict class labels)
Loss	Mean Squared Error (MSE)	Binary Cross-Entropy Loss
Activation	None (identity)	Sigmoid function $\sigma(z) = \frac{1}{1+e^{-z}}$

Part (c) — Role of the Sigmoid Function (2 Marks)

- The sigmoid function **squashes** the linear combination of inputs $z = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n$ to the range $(0, 1)$:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

- Its output can be interpreted as a **probability** ($P(\text{Pass})$). Values close to 1 indicate high confidence for the positive class; close to 0 indicates the negative class.
- This enables a **probabilistic interpretation** of the classification boundary.

Part (d) — Predicted Class for $p = 0.82$, Threshold = 0.5 (1 Mark)

Since $0.82 > 0.5$ (threshold), the model predicts: **Likely to Pass**.

Justification: Any output probability exceeding the decision threshold 0.5 is mapped to the positive class label ("Pass").

Part (e) — Limitation of Accuracy Alone (1 Mark)

Accuracy is **misleading for imbalanced datasets**. If 95% of students historically pass, a model predicting "Pass" for every student achieves 95% accuracy while completely failing to identify at-risk "Fail" students — the **Accuracy Paradox**. Metrics like **Precision**, **Recall**, and **F1-Score** must supplement accuracy.

UNIT IV: CCE2 — MODEL ANSWER KEY

Unsupervised Learning & Advanced ML Techniques

Course Pattern: TE IT - 2024 Pattern (SPPU)

Document Type: CCE2 Model Answer Key

Academic Term: Semester V (2026)

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Q1. Customer Segmentation using K-Means — 8 Marks

SCENARIO:

A retail company segments customers by Annual Income and Spending Score. Initial centroids: $C_1 = (2, 3)$, $C_2 = (8, 7)$.

Customer	Income	Spending Score
C1	2	3
C2	3	2
C3	4	3
C4	8	7
C5	9	8
C6	8	9

Table 1.1: Customer Dataset

Part (a) — Objective of K-Means Clustering (1 Mark)

K-Means clustering partitions a dataset into **K distinct, non-overlapping clusters** by **minimizing the within-cluster sum of squared distances** (intra-cluster variance) between each data point and its assigned centroid, while maximizing inter-cluster separation.

Part (b) — Euclidean Distances from Both Centroids (3 Marks)

Distance formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ Centroids: $C_1 = (2, 3)$, $C_2 = (8, 7)$

Cust.	Distance from $C_1(2,3)$	$d(C_1)$	Distance from $C_2(8,7)$	$d(C_2)$
C1(2,3)	$\sqrt{(2-2)^2 + (3-3)^2}$	0.00	$\sqrt{(2-8)^2 + (3-7)^2} = \sqrt{52}$	≈ 7.21
C2(3,2)	$\sqrt{(3-2)^2 + (2-3)^2} = \sqrt{2}$	≈ 1.41	$\sqrt{(3-8)^2 + (2-7)^2} = \sqrt{50}$	≈ 7.07
C3(4,3)	$\sqrt{(4-2)^2 + (3-3)^2} = \sqrt{4}$	2.00	$\sqrt{(4-8)^2 + (3-7)^2} = \sqrt{32}$	≈ 5.66
C4(8,7)	$\sqrt{(8-2)^2 + (7-3)^2} = \sqrt{52}$	≈ 7.21	$\sqrt{(8-8)^2 + (7-7)^2}$	0.00
C5(9,8)	$\sqrt{(9-2)^2 + (8-3)^2} = \sqrt{74}$	≈ 8.60	$\sqrt{(9-8)^2 + (8-7)^2} = \sqrt{2}$	≈ 1.41
C6(8,9)	$\sqrt{(8-2)^2 + (9-3)^2} = \sqrt{72}$	≈ 8.49	$\sqrt{(8-8)^2 + (9-7)^2} = \sqrt{4}$	2.00

Part (c) — Cluster Assignments (2 Marks)

Assign each customer to the centroid with the **minimum distance**:

Customer	$d(C_1)$	$d(C_2)$	Assigned Cluster
C1	0.00	7.21	Cluster 1
C2	1.41	7.07	Cluster 1
C3	2.00	5.66	Cluster 1
C4	7.21	0.00	Cluster 2
C5	8.60	1.41	Cluster 2
C6	8.49	2.00	Cluster 2

Result: Cluster 1 = {C1, C2, C3}, Cluster 2 = {C4, C5, C6}

Part (d) — New Centroid Calculation (1 Mark)

After assignment, new centroids are computed as the **mean of all data points** in each cluster:

$$C_1^{\text{new}} = \left(\frac{2+3+4}{3}, \frac{3+2+3}{3} \right) = (3, 2.67)$$

$$C_2^{\text{new}} = \left(\frac{8+9+8}{3}, \frac{7+8+9}{3} \right) = (8.33, 8)$$

The algorithm then re-assigns points to the new centroids and repeats until convergence.

Part (e) — Limitation of K-Means (1 Mark)

K-Means requires the number of clusters K to be **specified in advance**. It is also sensitive to **initial centroid placement**, susceptible to **outliers** skewing centroids, and cannot detect clusters of arbitrary or non-spherical shapes.

Q2. Market Basket Analysis — 8 Marks

SCENARIO:

A supermarket records 5 transactions. Apply Association Rule Mining to find frequent itemsets.

Transaction	Items
T1	Bread, Milk
T2	Bread, Milk, Butter
T3	Milk, Butter
T4	Bread, Milk, Butter
T5	Bread, Butter

Table 2.1: Transaction Database

Part (a) — Type of Unsupervised Technique (1 Mark)

Association Rule Mining (also known as Association Rule Learning). Example algorithm: **Apriori** or **FP-Growth**.

Part (b) — Support of {Bread, Milk} (2 Marks)

$$\text{Support}(\{A\}) = \frac{\text{Transactions containing } A}{\text{Total Transactions}}$$

Transactions containing **both** Bread and Milk: T1, T2, T4 → Count = 3. Total = 5.

$$\text{Support}(\{\text{Bread, Milk}\}) = \frac{3}{5} = 0.6 \quad (60\%)$$

Part (c) — Confidence of Bread → Milk (2 Marks)

$$\text{Confidence}(A \rightarrow B) = \frac{\text{Support}(A \cup B)}{\text{Support}(A)}$$

Transactions containing **Bread**: T1, T2, T4, T5 → Count = 4. Transactions containing **both Bread and Milk**: T1, T2, T4 → Count = 3.

$$\text{Confidence}(\text{Bread} \rightarrow \text{Milk}) = \frac{3}{4} = 0.75 \quad (75\%)$$

Part (d) — Confidence from Given Support Values (1 Mark)

Given: $\text{Support}(\text{Bread}) = \frac{4}{5}$, $\text{Support}(\text{Bread, Milk}) = \frac{3}{5}$

$$\text{Confidence} = \frac{\frac{3}{5}}{\frac{4}{5}} = \frac{3}{4} = 0.75$$

Part (e) — What High Confidence Indicates (1 Mark)

High confidence indicates a **strong association rule**: if a customer purchases the antecedent (Bread), it is highly likely they will also purchase the consequent (Milk). It reflects the **conditional probability** of the consequent given the antecedent.

Part (f) — Practical Application (1 Mark)

Cross-Selling and Product Placement: Retailers use association rules to place frequently co-purchased items (e.g., Bread near Milk) in adjacent shelf sections, or to generate “Customers also bought...” recommendations in e-commerce.

Q3. DBSCAN for Identifying Outliers — 7 Marks

SCENARIO:

A logistics company groups delivery locations by GPS coordinates. Some locations are isolated outliers.

Part (a) — Why DBSCAN is Useful (1 Mark)

DBSCAN (**Density-Based Spatial Clustering of Applications with Noise**) discovers clusters of **arbitrary shapes** in geospatial data and inherently identifies **noise points** (isolated delivery locations) as outliers without requiring a pre-specified number of clusters, making it ideal for this application.

Part (b) — Meaning of epsilon and MinPts (2 Marks)

- **epsilon (neighborhood radius):** Defines the maximum radius around a data point to search for neighboring points. Two points are “density-connected” if their distance is $\leq \epsilon$.
- **MinPts (minimum points):** The minimum number of points (including the point itself) required within the ϵ -neighborhood for a point to qualify as a **core point** and anchor a dense cluster region.

These two hyperparameters together define what constitutes a **dense region**.

Part (c) — Core, Border, and Noise Points (3 Marks)

Point Type	Definition	In this Context
Core Point	Has $\geq$ MinPts points within ϵ -radius (including itself)	A delivery cluster hub with many nearby stops
Border Point	Has $<$ MinPts neighbors but lies within ϵ -radius of a core point	A stop near a hub but not itself a hub
Noise Point	Neither a core point nor a border point; completely isolated	Remote, unusual delivery location — flagged as outlier

Part (d) — Advantage of DBSCAN over K-Means (1 Mark)

DBSCAN **does not require the number of clusters K to be specified in advance**; it discovers cluster count automatically from the data's density structure. It also handles **arbitrary cluster shapes** and explicitly identifies **outliers**, which K-Means cannot do.

Q4. Hierarchical Clustering for Customer Groups — 7 Marks

SCENARIO:

A company clusters customers without specifying the number of clusters in advance.

Part (a) — Basic Idea of Hierarchical Clustering (1 Mark)

Hierarchical clustering builds a **tree-like nested structure of clusters** (a **dendrogram**) by either iteratively merging individual data points into larger clusters (bottom-up) or recursively splitting one large cluster (top-down), without requiring K to be pre-defined.

Part (b) — Agglomerative vs Divisive (2 Marks)

Parameter	Agglomerative (Bottom-Up)	Divisive (Top-Down)
Start	Each point = its own cluster	All points in one cluster
Process	Iteratively merge two closest clusters	Iteratively split most heterogeneous cluster
Approach	AGNES (Agglomerative Nesting)	DIANA (Divisive Analysis)
Common Use	More widely used in practice	Less common; computationally expensive

Part (c) — Two Linkage Methods (2 Marks)

- Single Linkage (Min):** Distance between two clusters = **minimum** distance between any point in cluster A and any point in cluster B. Tends to produce elongated, “chaining” clusters.
- Complete Linkage (Max):** Distance between two clusters = **maximum** distance between any point in cluster A and any point in cluster B. Produces more compact, balanced clusters.

(Also acceptable: Average Linkage, Ward's Method)

Part (d) — Using a Dendrogram to Determine Cluster Count (2 Marks)

1. The **dendrogram** displays all merges as a tree; the **y-axis** represents the distance (dissimilarity) at which clusters were merged.
2. To determine the number of clusters: draw a **horizontal cut line** across the dendrogram. The number of **vertical branches** the line intersects equals the number of clusters.
3. The optimal cut is typically placed at the **largest vertical gap** (longest branch without a merge), indicating the most significant dissimilarity jump between cluster levels.

Q5. PCA for Dimensionality Reduction — 8 Marks

SCENARIO:

A dataset contains 20 correlated numerical features. Many are redundant, increasing computation cost and visualization difficulty.

Part (a) — Why Dimensionality Reduction is Required (2 Marks)

1. **Curse of Dimensionality:** As the number of features increases, the data becomes increasingly sparse in high-dimensional space, degrading model performance and increasing training time.
2. **Multicollinearity:** Correlated features add **redundant information**; reducing dimensions removes this redundancy without significant information loss.
3. **Visualization:** Datasets with 20 features cannot be visualized directly; reducing to 2-3 principal components enables 2D/3D plots for exploratory analysis.

Part (b) — Basic Objective of PCA (2 Marks)

PCA transforms a high-dimensional dataset into a **lower-dimensional space** (subspace of uncorrelated axes called **Principal Components**) while retaining the **maximum possible variance** from the original data. Principal components are ordered by decreasing explained variance, so the first few capture most of the information.

Part (c) — Major Steps in Applying PCA (3 Marks)

1. **Standardization:** Scale each feature to zero mean and unit variance (z -score normalization) so features with larger ranges do not dominate.
2. **Covariance Matrix Computation:** Compute the covariance matrix to capture pairwise linear relationships between all features.
3. **Eigen Decomposition:** Calculate the **eigenvectors** (principal component directions) and **eigenvalues** (variance explained) of the covariance matrix.
4. **Component Selection:** Sort eigenvectors by descending eigenvalue; select the top k components that together explain the desired proportion of total variance (e.g., 95%).
5. **Projection:** Multiply the original data matrix by the selected eigenvector matrix to project data onto the new k -dimensional principal component space.

Part (d) — PCA vs Feature Selection (1 Mark)

- **Feature Selection** retains a **subset of original features** (columns), preserving their interpretability and physical meaning.
- **PCA (Feature Extraction)** creates **entirely new synthetic features** (principal components) as linear combinations of originals — the new axes have no direct physical interpretation.

Q6. Feature Selection for a Predictive Model — 7 Marks**SCENARIO:**

A healthcare dataset has 50 features; only a subset is relevant for predicting patient risk category.

Part (a) — Purpose of Feature Selection (1 Mark)

Feature selection identifies and retains only the **most informative features**, eliminating irrelevant and redundant ones. This improves model **accuracy**, **interpretability**, **training speed**, and reduces the risk of **overfitting**.

Part (b) — Feature Selection vs Feature Extraction (2 Marks)

Parameter	Feature Selection	Feature Extraction
Output	Subset of original features	New derived features (transformations)
Interpretability	Original meaning preserved	Loses original physical meaning
Example	Select 10 most correlated features from 50	PCA projecting 50 features into 5 principal components

Part (c) — Two Feature Selection Techniques (2 Marks)

1. **Filter Methods:** Rank features using statistical measures (e.g., **Pearson Correlation**, **Chi-Square Test**, **Mutual Information**) independently of any model. Fast and computationally efficient.
2. **Wrapper Methods:** Evaluate subsets of features by training a model on each subset and comparing performance (e.g., **Recursive Feature Elimination (RFE)**, Forward/Backward Selection). More accurate but computationally expensive.

(Also acceptable: Embedded methods — Lasso L1 regularization, tree-based feature importance)

Part (d) — Effect on Model Performance (2 Marks)

- **Improved Generalization:** Removing irrelevant features reduces noise, preventing the model from learning spurious correlations — decreasing **overfitting** and improving performance on unseen data.
- **Faster Training:** Fewer features mean smaller input dimensionality, reducing computation time for both training and inference.
- **Better Interpretability:** A model with fewer, relevant features is easier to explain to clinicians for medical applications.

Q7. Comparing K-Means and DBSCAN — 8 Marks

SCENARIO:

A city-planning department identifies dense service-request clusters. Cluster count is unknown; some requests are isolated.

Part (a) — Why K-Means May Not Be Suitable (2 Marks)

1. K-Means requires the **number of clusters K to be predetermined**. Since the city planners do not know K in advance, an incorrect K will force data into meaningless groups.
2. K-Means assigns **every point to a cluster** — isolated service requests will distort centroids rather than being flagged as anomalies, degrading cluster quality.
3. K-Means assumes **spherical clusters of similar size**, which may not reflect the irregular geographic distribution of service requests.

Part (b) — How DBSCAN Handles Noise (2 Marks)

DBSCAN classifies points that have **fewer than MinPts neighbors within their ϵ -neighborhood** and are **not reachable from any core point** as **noise points (outliers)**. These points are explicitly labeled as -1 (unassigned) and are excluded from all clusters, cleanly separating isolated requests from dense service clusters.

Part (c) — Comparison: K-Means vs DBSCAN (3 Marks)

Parameter	K-Means	DBSCAN
Cluster Shape	Assumes spherical, convex clusters	Arbitrary shapes (concave, elongated, irregular)
No. of Clusters	Must be specified as K before running	Determined automatically from data density
Noise Handling	Forces all points into clusters; no outlier detection	Explicitly labels outlier points as noise (-1)
Sensitivity	Sensitive to initial centroid positions	Sensitive to ϵ and MinPts hyperparameters
Scalability	Scales well to large datasets ($O(nKt)$)	Slower for very high-dimensional data

Part (d) — Appropriate Technique and Justification (1 Mark)

DBSCAN is the appropriate technique.

Justification: The number of cluster groups is unknown, some service requests are isolated (outliers), and cluster shapes may be irregular (geographic regions). DBSCAN handles all three requirements — automatic cluster discovery, noise flagging, and arbitrary shape detection — without needing K in advance.

Q8. Cluster Evaluation — 7 Marks

SCENARIO:

A data analyst runs K-Means for different values of K and needs to evaluate which clustering is best.

Part (a) — Why Cluster Evaluation is Required (1 Mark)

Unsupervised learning has **no ground-truth labels** to compare against. Cluster evaluation metrics are necessary to objectively measure **cluster quality**, determine the **optimal number of clusters**, and verify that the algorithm found **meaningful, well-separated groupings** rather than arbitrary partitions.

Part (b) — Concept of the Silhouette Score (2 Marks)

The **Silhouette Score** measures how well each point fits within its assigned cluster compared to the nearest alternative cluster:

$$S(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

Where:

- $a(i)$ = mean distance from point i to all other points **in its own cluster** (cohesion — lower is better)
- $b(i)$ = mean distance from point i to all points **in the nearest other cluster** (separation — higher is better)

Score ranges from -1 to $+1$: closer to $+1$ indicates well-separated clusters; near 0 indicates overlapping; negative values indicate misclassification.

Part (c) — Interpretation of Scores 0.78 and 0.31 (2 Marks)

Score	Interpretation	Cluster Quality
0.78	Points are well inside their cluster and far from neighboring clusters	Strong, well-separated clusters
0.31	Points are close to the boundary between clusters; significant overlap exists	Weak cluster structure; overlapping clusters

The model with score **0.78** is **significantly superior** and should be preferred for the final customer segmentation.

Part (d) — Limitation of a Single Evaluation Metric (2 Marks)

A single metric can be **biased toward specific cluster geometries**:

- The Silhouette Score **favours spherical, convex clusters** (those formed by K-Means). Irregular but valid DBSCAN clusters may receive poor Silhouette scores even if the clustering is correct.
- Relying on one metric may lead to **selecting a suboptimal algorithm** or K value. It is best practice to use **multiple complementary metrics** (e.g., Silhouette Score + Davies-Bouldin Index + Calinski-Harabasz Index) and domain knowledge together for robust evaluation.

CCE2 — INTEGRATED CASE STUDIES

Additional Integrated Case-Study Questions — Model Answer Key

Course Pattern: TE IT - 2024 Pattern (SPPU)

Academic Term: Semester V (2026)

Document Type: CCE2 Model Answer Key

Course Code: PCC-302-IT

IMPORTANT:

These integrated case studies test cross-unit knowledge spanning **Unit III (Supervised Learning)** and **Unit IV (Unsupervised Learning)**. Solutions are structured point-wise per SPPU Model Answer Key guidelines. Award marks proportionally for correct reasoning even if final values differ.

Q1. E-Commerce Customer Analytics — 8 Marks

SCENARIO:

An e-commerce company wants to (1) predict whether a customer will purchase a product, and (2) segment customers by purchasing behaviour.

Part (a) — Supervised Learning Approach for Purchase Prediction (1 Mark)

Logistic Regression (binary classification: Purchase / Not Purchase). Also acceptable: **Decision Tree**, **Random Forest Classifier**, **Support Vector Machine (SVM)**.

Part (b) — Unsupervised Technique for Customer Segmentation (1 Mark)

K-Means Clustering — groups customers into distinct segments based on similarity in purchasing behaviour features without predefined labels. Also acceptable: **Hierarchical Clustering**, **DBSCAN**.

Part (c) — Why Training and Testing Datasets are Required (2 Marks)

- **Training Dataset:** The model learns the patterns between customer features (e.g., browsing history, demographic data) and the purchase outcome. Internal parameters (weights) are adjusted to minimize training error.
- **Testing Dataset:** Evaluates the trained model on **completely unseen data** to measure real-world **generalization performance**. Without a separate test set, we cannot detect overfitting — a model that memorizes training data but fails on new customers.

Part (d) — How K-Means is Applied to Customer Segmentation (2 Marks)

- 1. **Initialization:** Choose K segments and randomly initialize K cluster centroids in the feature space (e.g., spending score, visit frequency, average order value).
- 2. **Assignment:** Compute Euclidean distance from each customer’s feature vector to all K centroids; assign each customer to the **nearest centroid’s cluster**.
- 3. **Update:** Recalculate each centroid as the **mean** of all customer feature vectors in that cluster.
- 4. **Convergence:** Repeat assignment and update until centroid positions stabilize (change < threshold), producing K distinct customer segments for targeted marketing.

Part (e) — Two Evaluation Measures for the Classification Model (2 Marks)

Metric	Formula	Why Useful Here
Precision	$\frac{TP}{TP + FP}$	Reduces false alarms — avoids wasting promotion budget on non-buyers
F1-Score	$2 \times \text{Precision} \times \frac{\text{Recall}}{\text{Precision} + \text{Recall}}$	Balances both metrics; ideal when buyer class is the minority (imbalanced dataset)

Q2. Healthcare Machine Learning Pipeline — 8 Marks

SCENARIO:

A hospital predicts **high-risk** vs **low-risk** patients. Dataset has missing values, categorical attributes, and numerical features with different ranges.

Part (a) — Two Preprocessing Operations (1 Mark)

- 1. **Missing Value Imputation:** Fill missing numerical values with the column mean/median; fill missing categorical values with the mode. Prevents model training errors.
- 2. **Categorical Encoding:** Convert categorical attributes (e.g., Gender, Blood Type) to numerical format using **One-Hot Encoding** or **Label Encoding**, since ML algorithms require numerical input.

Part (b) — Purpose of Scaling Numerical Features (1 Mark)

Healthcare features like Age (0–100), Blood Pressure (60–180), and Cholesterol (100–400) have **vastly different numerical ranges**. Scaling (e.g., **Min-Max Normalization** or **Z-score Standardization**) brings all features to a uniform range, preventing features with larger magnitudes from **dominating distance calculations** (in KNN) or **gradient updates** (in Logistic Regression/Neural Networks).

Part (c) — How KNN is Used for Patient Classification (2 Marks)

1. **Define k :** Choose the number of nearest neighbors k (e.g., $k = 5$).
2. **Distance Computation:** For a new, unclassified patient, compute the Euclidean distance between the patient's feature vector and every training patient.
3. **Neighbor Identification:** Select the k training patients with the **smallest distances** (nearest neighbors).
4. **Majority Voting:** The new patient is assigned the class (**High Risk** or **Low Risk**) that appears **most frequently** among the k neighbors.

Part (d) — Calculate Precision and Recall (2 Marks)

Given: TP = 85, TN = 90, FP = 15, FN = 10

Precision (What fraction of predicted high-risk patients are truly high-risk?):

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{85}{85 + 15} = \frac{85}{100} = 0.85 \text{ (85\%)}$$

Recall (What fraction of all actual high-risk patients were correctly identified?):

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{85}{85 + 10} = \frac{85}{95} \approx 0.8947 \text{ (89.47\%)}$$

Part (e) — Precision or Recall for High-Risk Patient Identification? (2 Marks)

Recall should receive greater attention.

- **Reasoning:** Failing to identify a high-risk patient = **False Negative** — the patient receives no treatment and may suffer serious consequences (e.g., death, permanent injury).
- **Recall** = $\frac{TP}{TP + FN}$ — maximizing Recall **minimizes False Negatives**, ensuring the fewest high-risk patients are missed.
- A False Positive (low-risk patient incorrectly flagged as high-risk) reduces Precision but only leads to additional (unnecessary) tests — a far less severe consequence than missing a genuinely high-risk case.

Q3. Retail Intelligence Case Study — 7 Marks**SCENARIO:**

A supermarket wants to (1) group customers by purchasing behaviour, and (2) discover products frequently purchased together.

Part (a) — Unsupervised Technique for Customer Grouping (1 Mark)

K-Means Clustering (or Hierarchical Clustering) — groups customers into distinct segments based on purchasing behaviour without predefined labels.

Part (b) — Technique for Frequent Product Combinations (1 Mark)

Association Rule Mining — specifically the **Apriori algorithm** or **FP-Growth algorithm** — discovers frequent itemsets and generates rules like $\{\text{Bread}\} \rightarrow \{\text{Butter}\}$.

Part (c) — How K-Means is Used for Customer Segmentation (2 Marks)

1. **Select K** : Define the desired number of customer segments (e.g., $K = 3$: budget shoppers, premium buyers, occasional buyers).
2. **Initialize Centroids**: Randomly place K centroids in the feature space (e.g., total spend, visit frequency, product category diversity).
3. **Assign Customers**: Compute Euclidean distance from each customer to each centroid; assign customer to the **nearest centroid's cluster**.
4. **Update Centroids**: Recalculate each centroid as the **mean** of all customers in that cluster.
5. **Iterate**: Repeat until centroids stabilize, producing distinct customer segments for targeted promotions.

Part (d) — Support and Confidence in Association Rule Mining (2 Marks)

- **Support**: Measures the **frequency** of an itemset appearing across all transactions. Indicates how popular an item/combination is.

$$\text{Support}(A \rightarrow B) = \frac{\text{Count of transactions containing both A and B}}{\text{Total transactions}}$$

- **Confidence**: Measures the **conditional probability** that B is purchased given A is purchased. Indicates rule strength.

$$\text{Confidence}(A \rightarrow B) = \frac{\text{Support}(A \cup B)}{\text{Support}(A)} = \frac{\text{Count}(A \text{ and } B)}{\text{Count}(A)}$$

Example: If $\text{Support}(\{\text{Bread}, \text{Butter}\}) = 0.6$ and $\text{Support}(\{\text{Bread}\}) = 0.8$, then $\text{Confidence}(\text{Bread} \rightarrow \text{Butter}) = 0.6/0.8 = 0.75$.

Part (e) — Business Decision Supported by Results (1 Mark)

Cross-Selling & Store Layout Optimization: Place products identified as co-purchased (via Association Rules) in adjacent shelf sections. Direct targeted discount offers to the relevant customer segments identified by K-Means — e.g., offering Butter coupons to the “Bread buyers” cluster — increasing sales volume and customer satisfaction.

Q4. End-to-End Machine Learning Scenario — 8 Marks

SCENARIO:

A startup predicts whether an online customer will **cancel their subscription**. Dataset has missing values, categorical attributes, and numerical features.

Part (a) — Arrange ML Workflow Stages (2 Marks)

Correct order of the Machine Learning workflow:

Step	Stage	Purpose
1	Data Collection	Gather raw customer subscription data from databases / logs
2	Data Preprocessing	Handle missing values, encode categoricals, scale numericals
3	Training	Fit the ML model on the training partition to learn patterns
4	Validation	Tune hyperparameters; monitor for overfitting on held-out data
5	Testing	Final unbiased performance evaluation on completely unseen data

Part (b) — Why Divide into Training, Validation, and Testing Sets (2 Marks)

- **Training Set:** Used to **fit the model** — adjust weights, learn decision boundaries, minimize training loss. The model “sees” this data repeatedly.
- **Validation Set:** Used **during development** for **hyperparameter tuning** (e.g., optimal regularization, tree depth) without contaminating the final test evaluation. Provides early detection of overfitting.
- **Testing Set:** Used **only once** at the very end. Provides a **final unbiased estimate** of real-world generalization — simulates how the model performs on new, unseen subscription data. Using training or validation data for final evaluation would give an **overly optimistic** performance estimate.

Part (c) — Metric for Many False Negatives (2 Marks)

Recall (also called **Sensitivity** or **True Positive Rate**) should be examined carefully.

- **Definition:** $\text{Recall} = \frac{TP}{TP + FN}$
- **Reasoning:** A False Negative here means the model **failed to predict** that a customer will cancel — so no retention action is taken, and the subscription is lost.
- A high FN count directly **lowers Recall**. By monitoring and improving Recall, the startup ensures it catches as many at-risk customers as possible before they churn, enabling timely retention campaigns.

Part (d) — Unsupervised Technique for Grouping Without Labels (2 Marks)

Technique: K-Means Clustering

Purpose: K-Means groups customers into K clusters based on feature similarity (usage frequency, payment history, subscription tier, support tickets) **without requiring predefined class labels** like “cancelled” or “retained.” This reveals hidden **behavioural segments** (e.g., light users, power users, dormant accounts), enabling the startup to tailor retention strategies for each segment proactively — before cancellation occurs.